

Total marks – 120**Attempt All Questions****All questions are of equal value****Answer each question in a SEPARATE writing booklet. Extra booklets are available.****Question 1 (15 Marks)****Marks**

a) Find $\int \frac{1}{\sqrt{x^2+9}} dx$. 1

b) Use integration by parts to evaluate $\int_1^e \frac{\ln x}{\sqrt{x}} dx$ 3

c) Using the substitution $u = 1 - x$ evaluate $\int_0^{\frac{1}{2}} \frac{x}{(1-x)^2} dx$ 3

d) Find $\int \frac{dx}{x^2 + 4x + 7}$. 2

e) (i) Show, using a suitable substitution that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{\cos x}{\cos x + \sin x} dx$. 4

Question 2 (15 Marks) Use a SEPARATE writing booklet.**Marks**

a) Let $z = \frac{7-i}{3-4i}$.

(i) Find $|z|$. 2

(ii) Evaluate $\tan \left\{ \tan^{-1} \left(\frac{4}{3} \right) - \tan^{-1} \left(\frac{1}{7} \right) \right\}$. 2

(iii) Hence find the principal argument of $\frac{7-i}{3-4i}$ in terms of π . 2

b) The point P represents the complex number z on the Argand diagram. Describe the locus of P when $\arg(z-2) = \arg(z+2) + \frac{\pi}{2}$. 2

c) (i) Assuming the result $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$, and using a suitable substitution, solve the equation $8x^3 - 6x + 1 = 0$. 3

(ii) Hence find the value of

$\alpha) \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9}$. 1

$\beta) \sec \frac{2\pi}{9} + \sec \frac{4\pi}{9} + \sec \frac{8\pi}{9}$. 3

Question 3 (15 Marks) Use a SEPARATE writing booklet.**Marks**

- a) Sketch the functions $g(x) = \sqrt{9 - x^2}$ and $h(x) = x$ on the same axes. 3

Use these graphs to sketch $y = f(x)$ where $f(x) = g(x).h(x)$. Hence sketch each of the following on separate number planes.

(i) $y = f(-x)$ 1

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = |f(x)|$ 1

(iv) $y^2 = f(x)$ 2

- b) (i) Show that $z = i$ is a root of the equation $(2 - i)z^2 - (1 + i)z + 1 = 0$. 1

- (ii) Find the other root of the equation in the form $z = a + ib$, where a and b are real numbers. 2

- c) Let p, q, r be the roots of the equation $x^3 - 4x + 7 = 0$. Write down the cubic equation in x whose roots are p^2, q^2 and r^2 . 3

Question 4 (15 Marks) Use a SEPARATE writing booklet.**Marks**

- a) A particle of mass 1 kg is projected vertically upwards under gravity with a speed of $2c$ in a medium which the resistance to motion is $\frac{g}{c^2}$ times the square of the speed, where c is positive constant.

- (i) Show that the maximum height (H) reached is

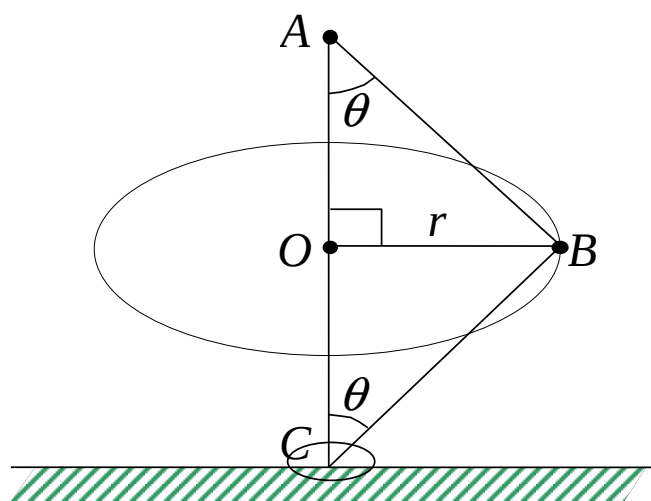
3

$$H = \frac{c^2}{2g} \ln 5.$$

- (ii) Show that the speed with which the particle returns to its starting point is given by $v = \frac{2c}{\sqrt{5}}$.

4

- b) Two light rigid rods AB and BC , each of length 0.5 m, are smoothly jointed at B and the rod is smoothly jointed at A to a fixed smooth vertical rod.



The joint at B has a particle of mass 2 kg attached. A small ring of mass 1 kg is smoothly jointed to BC at C and can slide on the vertical rod below A . The ring rests on a smooth horizontal ledge at a distance $\frac{\sqrt{3}}{2}$ m below A . The system rotates about the vertical rod with constant angular velocity 6 radians per second. Find:

- (i) the forces in the rod AB and BC ;

5

- (ii) the forces exerted by the ledge on the ring. (let $g = 10 \text{ m/s}^2$)

3

Question 5 (15 Marks) Use a **SEPARATE** writing booklet.**Marks**

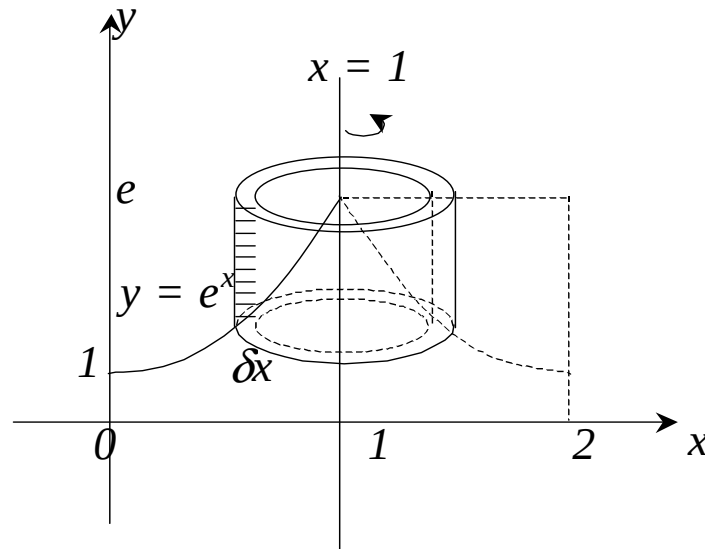
- a) i) Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ has the equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. **3**
- ii) This ellipse meets the y-axis at C and D . Tangents drawn at C and D on the ellipse meet the tangent in (i) at the points E, F respectively. Prove that $CE \cdot DF = a^2$. **4**
- b) i) Show that if $y = mx + k$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $m^2 a^2 - b^2 = k^2$. **3**
- ii) Hence find the equation of the tangents from the point $(1, 3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$ and the coordinates of their points of contact. **5**

End of Question 5.**Please Turn Over.**

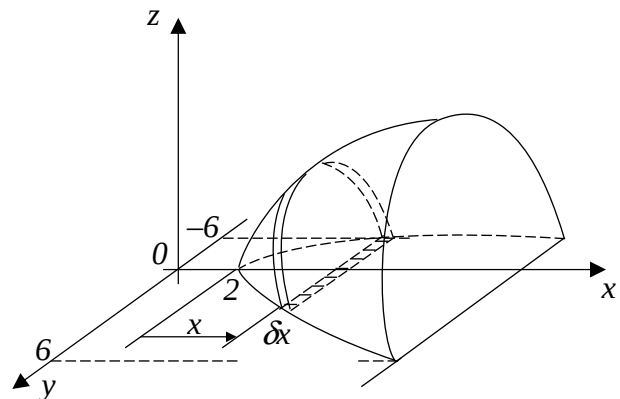
Question 6 (15 Marks) Use a SEPARATE writing booklet.

Marks

- a) By taking strips parallel to the axis of rotation, use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by $y = e^x$, $y = e$ and the y -axis about the line $x = 1$. **6**



- b) The base of a particular solid is the region bounded by the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$ between its vertex $(2, 0)$ and the corresponding latus rectum. Every cross-section perpendicular to the major axis is a semicircle with diameter in the base of the solid.

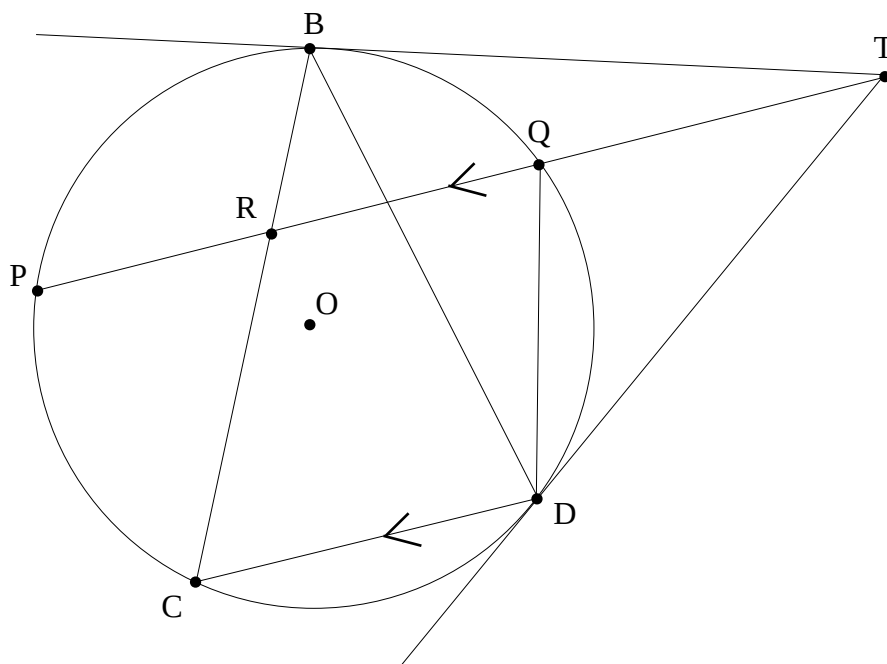


- i) Find the equation of the latus rectum. **2**
- ii) Find the volume of the solid. **4**

- c) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$. **3**
- The chord PQ subtends a right angle at another point $R\left(cr, \frac{c}{r}\right)$ on the hyperbola.
- Show that the normal at R is parallel to PQ .

Question 7 (15 Marks) Use a SEPARATE writing booklet.**Marks**

a)



PQ, CD are parallel chords of a circle, centre O. The tangent at D meets PQ extended at T. B is the point of contact of the other tangent from T. BC meets PQ at R.

- | | | |
|-------|--|---|
| (i) | Copy the diagram . | |
| (ii) | Prove that $\angle BDT = \angle BRT$ and hence state why B, T, D and R are concyclic points. | 3 |
| (iii) | Prove $\angle BRT = \angle DRT$. | 3 |
| (iv) | Show that $\triangle RCD$ is isosceles. | 2 |
| (v) | Prove that $\triangle PRC \equiv \triangle QRD$. | 3 |
-
- b) The equation $x^3 + 3px^2 + 3qx + r = 0$, where $p^2 \neq q$, has a double root. Show that
- $$4(p^2 - q)(q^2 - pr) = (pq - r)^2.$$
- 4

Question 8 (15 Marks) Use a SEPARATE writing booklet.**Marks**

- a) A coin is tossed six times. What is the probability that there will be more tails on the first three of the six throws than on the last three throws? **3**
- b) If m points are taken on a straight line and n points on a parallel line, how many triangles can be drawn each having its vertices at 3 of the given points? **3**
- c) (i) Show that $(1-x^2)^{\frac{n-3}{2}} - (1-x^2)^{\frac{n-1}{2}} = x^2(1-x^2)^{\frac{n-3}{2}}$. **1**
- (ii) Let $I_n = \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx$ where $n = 0, 1, 2, \dots$, **3**
- Show that $nI_n = (n-1)I_{n-2}$ for $n = 2, 3, 4, \dots$.
- (iii) Let $J_n = nI_n \cdot I_{n-1}$ for $n = 1, 2, 3, \dots$. **3**
- By using mathematical induction, prove that
- $$J_n = \frac{\pi}{2} \text{ for } n = 1, 2, 3, \dots$$
- (iv) Briefly explain why $0 < I_n < I_{n-1}$ for $n = 1, 2, 3, \dots$ **2**

END OF PAPER

Ex 2 Trial 2004.

$$a) \int \frac{1}{\sqrt{x^2+9}} dx = \ln x + \sqrt{x^2+9} + C \quad \checkmark$$

$$b) \int_1^e \frac{\ln x}{\sqrt{x}} dx = \int_1^e \ln x \frac{d}{dx}(2\sqrt{x}) dx \quad \checkmark$$

$$= \left[2 \ln x \sqrt{x} \right]_1^e - \int_1^e 2\sqrt{x} \cdot \frac{1}{x} dx \quad \checkmark$$

$$= 2\sqrt{e} - \left[4\sqrt{x} \right]_1^e \quad \checkmark$$

$$= 4 - 2\sqrt{e}.$$

$$c) \int_0^{\frac{1}{2}} \frac{x}{(1-x)^2} dx \quad \text{let } u = 1-x \therefore x = 1-u$$

$$dx = -du$$

$$x=0 \quad u=1$$

$$x=\frac{1}{2} \quad u=\frac{1}{2} \quad \checkmark$$

$$= \int_1^{\frac{1}{2}} \frac{(1-u)x - du}{u^2}$$

$$= \int_{\frac{1}{2}}^1 \left(\frac{1}{u^2} - \frac{1}{u} \right) du \quad \checkmark$$

$$= \left[-\frac{1}{u} - \ln u \right]_{\frac{1}{2}}^1$$

$$= -1 - 0 - (-2 - \ln \frac{1}{2})$$

$$= 1 - \ln 2. \quad \checkmark$$

$$1) \int \frac{dx}{x^2+4x+7} = \int \frac{dx}{(x+2)^2+3} \quad \checkmark$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+2}{\sqrt{3}} \right) + C. \quad \checkmark$$

$$c) i) \int_0^a f(x) dx \quad \text{let } x = a-u$$

$$dx = -du.$$

$$x=0 \Rightarrow u=a \quad \checkmark$$

$$x=a \Rightarrow u=0$$

$$= \int_a^0 (a-u)x - du$$

$$= \int_0^a (a-u) du = \int_0^a (a-u) dx \quad \checkmark$$

$$ii) \int_0^{\frac{\pi}{4}} \frac{\cos x dx}{\cos x + \sin x} = \int_0^{\frac{\pi}{4}} \frac{\cos(\frac{\pi}{4}-x) dx}{\cos(\frac{\pi}{4}-x) + \sin(\frac{\pi}{4}-x)} \quad \checkmark$$

$$= \int_0^{\frac{\pi}{4}} \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right) dx$$

$$\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \left(1 + \frac{\sin x}{\cos x} \right) dx \quad \checkmark$$

$$= \frac{1}{2} \left[x - \ln \cos x \right]_0^{\frac{\pi}{4}} \quad \checkmark$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \ln \frac{1}{\sqrt{2}} - (0 - 0) \right] \quad \checkmark$$

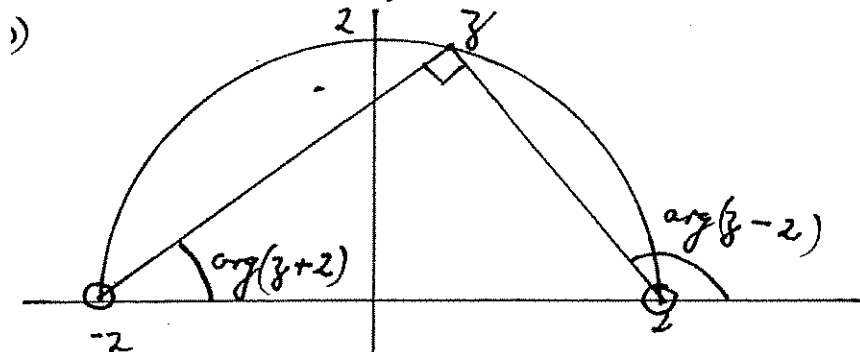
$$= \frac{\pi}{8} + \frac{1}{4} \ln 2.$$

1 a). $z = \frac{7-i}{3-4i} \times \frac{3+4i}{3+4i}$
 $= \frac{25+25i}{25} = 1+i.$

(i) $|z| = \sqrt{2} \neq \frac{|\sqrt{50}|}{|5|} = \sqrt{2}.$ ✓✓

ii) let $\alpha = \tan^{-1}(\frac{4}{3})$ $\beta = \tan^{-1}(\frac{1}{7})$
 $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
 $= \frac{\frac{4}{3} - \frac{1}{7}}{1 + \frac{4}{3} \times \frac{1}{7}}$
 $= 1.$ ✓

iii) since $z = 1+i$ principal arg $= \frac{\pi}{4}$ ✓✓
 $\arg\left(\frac{7-i}{3-4i}\right) = \arg(7-i) - \arg(3-4i)$
 $= \tan^{-1}(-\frac{1}{7}) - \tan^{-1}(-\frac{4}{3})$
 $= \tan^{-1}(\frac{4}{3}) - \tan^{-1}(\frac{1}{7})$
 $= \tan^{-1}(1) \text{ by (ii)}$
 $= \frac{\pi}{4}.$



clearly from diagram $\arg(z-2) = \arg(z+2) + \frac{\pi}{2}$
 since exterior angle of Δ equal sum of remote interior angles.

$\therefore$ locus of z is the semicircle shown with equation $y = \sqrt{4-x^2}$ for $y > 0$. ✓✓
 Note end points are excluded since $\arg 0$ is not defined.

c) (i) let $x = \cos \theta$

$\therefore 8x^3 - 6x + 1 = 0$

$\Rightarrow 2(4\cos^3 \theta - 3\cos \theta) = -1$

$\cos 3\theta = -\frac{1}{2}$ ✓

$3\theta = 2n\pi \pm \frac{2\pi}{3}.$ ✓

$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$ only 3 roots since cubic

$\therefore x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}.$ ✓

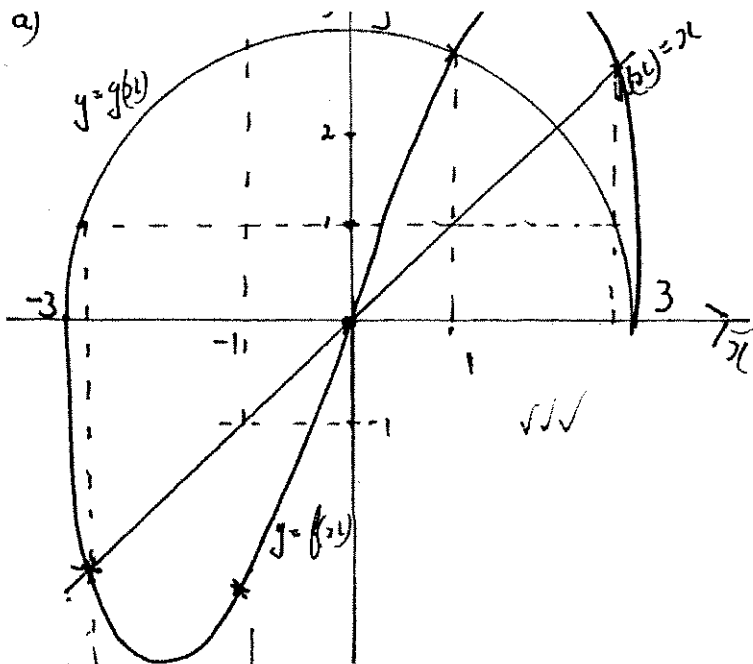
ii) $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = \sum \alpha = 0$ ✓

iii) $\sec \frac{2\pi}{9} + \sec \frac{4\pi}{9} + \sec \frac{8\pi}{9} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

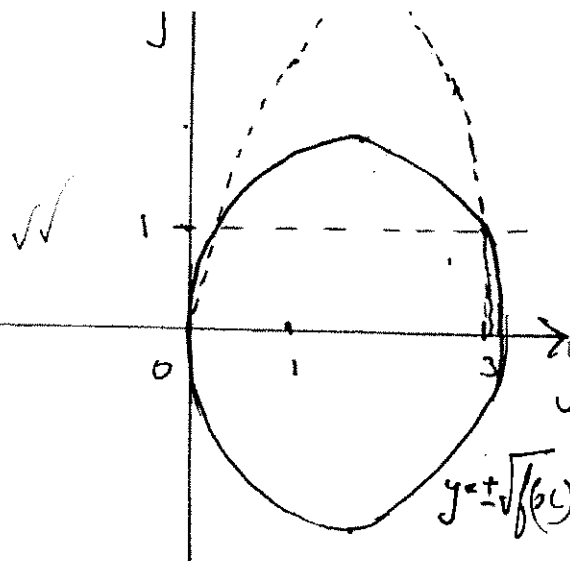
$= \frac{\sum \alpha \beta}{\alpha \beta \gamma}$ ✓

$= \frac{-6/8}{-1/4}$

$= 6.$ ✓



(iv)



b) $P(i) = (2-i)x - 1 - (1+i)i + 1$
 $= -2 + i - i + 1 + 1$
 $= 0 \therefore z = i$ is a root.

Product of roots: let other root be α .

$\alpha \times i = \frac{1}{2-i}$
 $= \frac{2+i}{5}$

$\times$ b.s by $-i$

$\alpha = \frac{1}{5} - \frac{2}{5}i$.

c) let $x = \sqrt{y}$.
 $y^{\frac{3}{2}} - 4y^{\frac{1}{2}} = -7$.

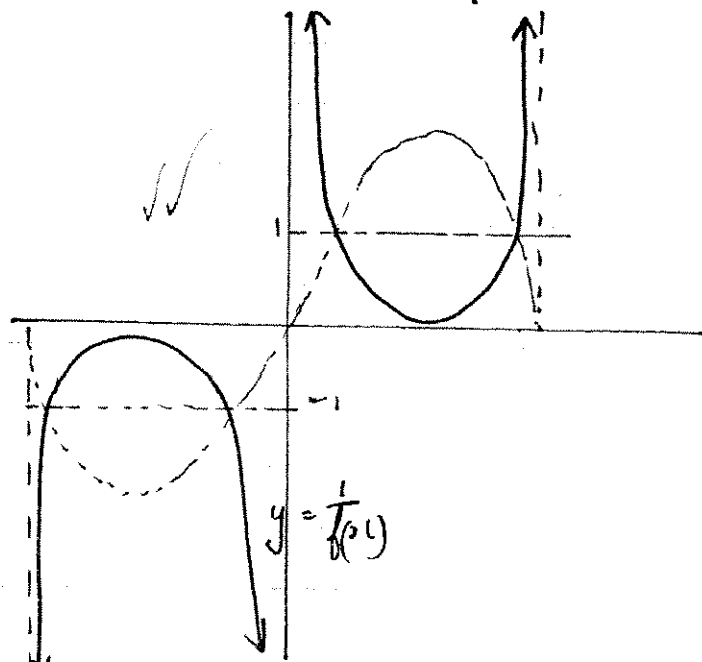
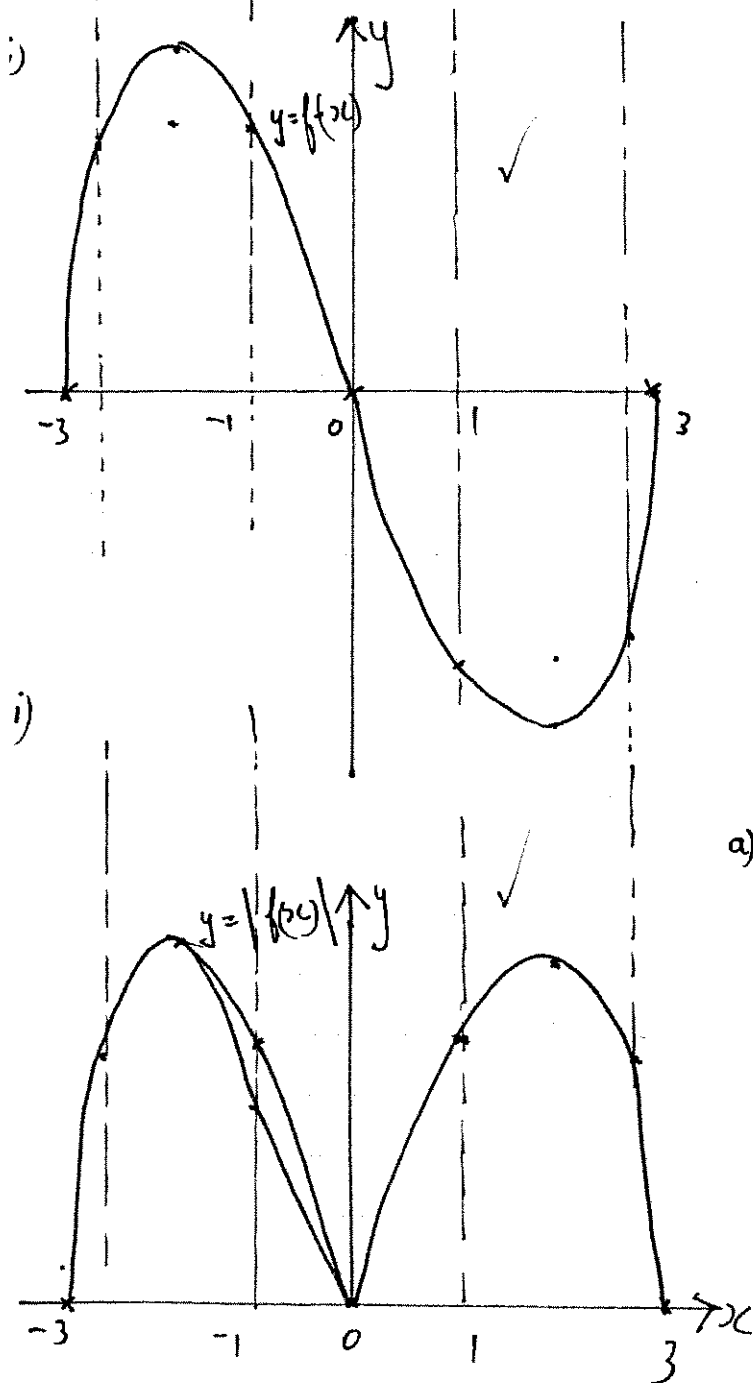
square b.s

$y^3 - 8y^2 + 16y = 49$.

Poly in x

$x^3 - 8x^2 + 16x - 49$.

a) ii)



Question 4 (15 Marks)

a)

A particle of mass 1 kg is projected vertically upwards under gravity with a speed of $2c$ in a medium which the resistance to motion is $\frac{g}{c^2}$ times the square of the speed, where c is positive constant.

(i) Show that the maximum height (H) reached is

$$H = \frac{c^2}{2g} \ln 5.$$

SOLUTION:

Upward motion. Choose a point of projection as origin and $\uparrow$ as positive.

Initial conditions: $t = 0, x = 0, v = 2c$.

Equation of motion: $\ddot{x} = -g - \frac{g}{c^2} v^2$.

Expression relating x and v :

$$v \frac{dv}{dx} = -g - \frac{g}{c^2} v^2,$$

$$-g dx = \frac{v dv}{1 + \frac{v^2}{c^2}},$$

$$-gx + A = \frac{c^2}{2} \ln \left(1 + \frac{v^2}{c^2}\right), A \text{ constant;}$$

$$x = 0, v = 2c$$

$$A = \frac{c^2}{2} \ln 5$$

$$x = \frac{c^2}{2g} \ln \frac{5c^2}{c^2 + v^2} \dots (1)$$

When the particle reaches its highest point, its velocity is zero.

So $v = 0$

from (2) $t = \frac{c \cdot \tan^{-1} 2}{g}$ is the time of ascent.

Let h be the distance between the point of projection and the highest point.

Then $v = 0$ from (1)

$$h = \frac{c^2}{2g} \ln 5.$$

Question 4 a) (ii)

Show that the speed with which the particle returns to its starting point is given by $v = \frac{2c}{\sqrt{5}}$.

SOLUTION:

Downward motion.

Origin at highest point and $\downarrow$ as positive direction.

Initial conditions: $t = 0, x = 0, v = 0$.

Equation of motion: $\ddot{x} = g - \frac{g}{c^2} v^2$.

Terminal velocity: as $\ddot{x} \rightarrow 0, v \rightarrow (c)^- \quad v < c$.

Expression relating x and v :

$$v \frac{dv}{dx} = g - \frac{g}{c^2} v^2$$

$$g dx = \frac{v dv}{1 - \frac{v^2}{c^2}}$$

$$gx + A = \frac{-c^2}{2} \ln \left(1 - \frac{v^2}{c^2} \right)$$

, A constant;

$$x = 0, v = 0$$

$$A = 0$$

$$x = \frac{c^2}{2g} \ln \frac{c^2}{c^2 - v^2} \quad \dots (2)$$

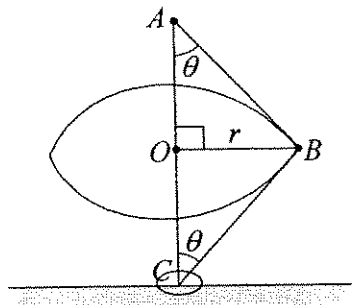
When the particle returns to its starting point, $x = h$.

$$\text{Hence from (2)} \quad h = \frac{c^2}{2g} \ln \frac{c^2}{c^2 - v^2}.$$

$$h = \frac{c^2}{2g} \ln 5$$

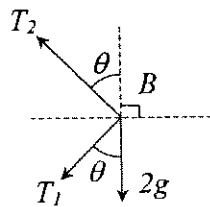
$$\text{But } 5 = \frac{c^2}{c^2 - v^2}$$

$$v = \frac{2c}{\sqrt{5}}$$

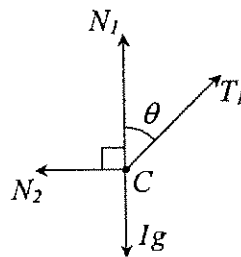
Question 4 b) (i)**SOLUTION:**

$$AB = BC = \frac{1}{2}, AC = \frac{\sqrt{3}}{2}.$$

Forces on B



Forces on C



N_2 is the force exerted by the rod AC on the ring C, and N_1 is the force exerted by the ledge.

The resultant force on B is $2\omega^2 r$ towards O.

The vertical component is zero

$$T_2 \cos \theta - T_1 \cos \theta = 2g \dots (1)$$

The horizontal component is $2\omega^2 r$

$$T_2 \sin \theta + T_1 \sin \theta = 2\omega^2 r \dots (2)$$

$$\text{But } \omega = 6, r = \sqrt{AB^2 - AO^2}$$

$$r = \frac{1}{4} \cos \theta = \frac{AO}{AB}$$

$$\cos \theta = \frac{\sqrt{3}}{2}, \sin \theta = \frac{1}{2}$$

Hence from (1) and (2) we obtain:

$$T_2 - T_1 = \frac{4g}{\sqrt{3}} \dots (3)$$

$$T_2 + T_1 = 36 \dots (4)$$

$$(3) + (4) T_2 = \frac{2g}{\sqrt{3}} + 18, T_2 = \frac{20}{\sqrt{3}} + 18 \text{ N ;}$$

$$(4) - (3)$$

$$T_1 = 18 - \frac{2g}{\sqrt{3}}$$

$$T_1 = 18 - \frac{20}{\sqrt{3}} \text{ N}$$

Question 4 b) (ii)

the forces exerted by the ledge on the ring. (let $g = 10\text{m/s}^2$)

The resultant force on C is zero.

For its vertical component we have

$$N_1 + T_1 \cos \theta = Ig$$

$$N_1 = g - \left(18 - \frac{20}{\sqrt{3}}\right) \frac{\sqrt{3}}{2}$$

$$N_1 = g + 10 - 9\sqrt{3}$$

$$N_1 = 20 - 9\sqrt{3} \text{ N}$$

Question 5 (15 Marks)

a) i)

Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ has the equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$.

SOLUTION:

This ellipse meets the y-axis at C and D . Tangents drawn at C and D on the ellipse meet the tangent in (i) at the points E , F respectively. Prove that $CE \cdot DF = a^2$.

SOLUTION:

(i)

Coordinates of C and D are $(0, b)$ and $(0, -b)$ respectively.

$\therefore$ the equations of the tangents through C and D are $y = b$ and $y = -b$, respectively.

Solve each of these equations simultaneously with the equation of the tangent at P ,

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ and we have the coordinates of E and F as follows:

$$E\left(\frac{a(1 - \sin \theta)}{\cos \theta}\right) \text{ and } F\left(\frac{a(1 + \sin \theta)}{\cos \theta}\right)$$

$$\begin{aligned} \therefore CE \cdot DF &= \left(\frac{a(1 - \sin \theta)}{\cos \theta}\right) \cdot \left(\frac{a(1 + \sin \theta)}{\cos \theta}\right) \\ &= \frac{a^2(1 - \sin^2 \theta)}{\cos^2 \theta} \\ &= \frac{a^2 \cos^2 \theta}{\cos^2 \theta} \\ &= a^2 \end{aligned}$$

b) i)

Show that if $y = mx + k$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $m^2 a^2 - b^2 = k^2$.

SOLUTION: The hyperbola has parametric equations $x = a \sec \theta$ and $y = b \tan \theta$. Hence

$$\frac{dy}{dx} = \frac{b \sec \theta}{a \tan \theta}.$$

If $y = mx + k$ is a tangent to the hyperbola at $P(a \sec \phi, b \tan \phi)$, then

$$m = \frac{dy}{dx} \text{ at } P \quad \dots (1)$$

$$ma \tan \phi - b \sec \phi = 0$$

P lies on $y = mx + k$

$$ma \sec \phi - b \tan \phi = -k \dots (2)$$

$$(2)^2 - (1)^2$$

$$m^2 a^2 (\sec^2 \phi - \tan^2 \phi) + b^2 (\tan^2 \phi - \sec^2 \phi) = k^2.$$

$$m^2 a^2 - b^2 = k^2$$

$$(2) \times \sec \phi - (1) \times \tan \phi$$

$$ma(\sec^2 \phi - \tan^2 \phi) = -k \sec \phi$$

$$a \sec \phi = -\frac{ma^2}{k},$$

$$(2) \times \tan \phi - (1) \times \sec \phi$$

$$b(\sec^2 \phi - \tan^2 \phi) = -k \tan \phi$$

$$b \tan \phi = -\frac{b^2}{k}$$

Therefore the point of contact of the tangent $y = mx + k$ is $P(-\frac{ma^2}{k}, -\frac{b^2}{k})$.

ii)

Hence find the equation of the tangents from the point $(1, 3)$ to the hyperbola

$$\frac{x^2}{4} - \frac{y^2}{15} = 1 \text{ and the coordinates of their points of contact.}$$

SOLUTION:

Now tangents from the point $(1, 3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$ have equations of the form $y - 3 = m(x - 1)$, that is, $y = mx + (3 - m)$.

$$\text{Hence } m^2 a^2 - b^2 = k^2$$

$$4m^2 - 15 = (3 - m)^2$$

$$3m^2 + 6m - 24 = 0$$

$$(m - 2)(m + 4) = 0.$$

$$\therefore m = 2,$$

$$k = 3 - m = 1$$

$$\text{and } P\left(-\frac{ma^2}{k}, -\frac{b^2}{k}\right) \equiv P(-8, -15)$$

$$\text{or } m = -4, k = 3 - m = 7 \text{ and}$$

$$P\left(-\frac{ma^2}{k}, -\frac{b^2}{k}\right) \equiv P\left(\frac{16}{7}, -\frac{15}{7}\right).$$

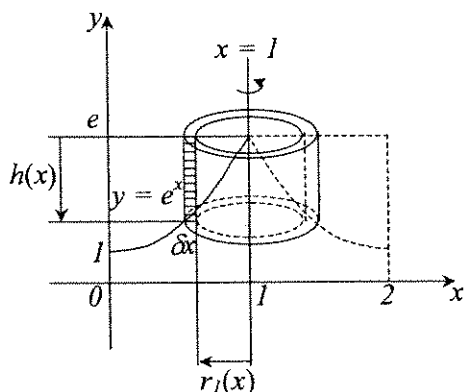
Hence the tangents from the point $(1, 3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$ are $y = 2x + 1$, with point of contact $P(-8, -15)$ and $y = -4x + 7$, with point of contact $P\left(\frac{16}{7}, -\frac{15}{7}\right)$.

End of Question 5.

Question 6 (15 Marks)

a)

By taking strips parallel to the axis of rotation, use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by $y = e^x$, $y = e$ and the y -axis about the line $x = 1$.

SOLUTION:

The typical cylindrical shell has radii $r_1(x) = 1 - x$, $r_2(x) = 1 - x + \delta x$, and height $h(x) = e - e^x$. This shell has volume

$$\begin{aligned}\delta V &= \pi[(1 - x + \delta x)^2 - (1 - x)^2]h(x) \\ &= 2\pi(1 - x)(e - e^x) \delta x \quad (\text{ignoring } (\delta x)^2).\end{aligned}$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi(1 - x)(e - e^x) \delta x$$

$$\therefore V = \int_0^1 2\pi(1 - x)(e - e^x) dx$$

$$= 2\pi \left[e \int_0^1 (1 - x) dx - \int_0^1 (1 - x) e^x dx \right]$$

$$= 2\pi \left[e \left(x - \frac{x^2}{2} \right) \Big|_0^1 - \int_0^1 (1 - x) de^x \right]$$

$$= 2\pi \left[\frac{e}{2} - ((1 - x)e^x) \Big|_0^1 - \int_0^1 (-1) \cdot e^x dx \right]$$

$$= 2\pi \left[\frac{e}{2} + 1 - e^x \Big|_0^1 \right]$$

$$= \pi(4 - e)$$

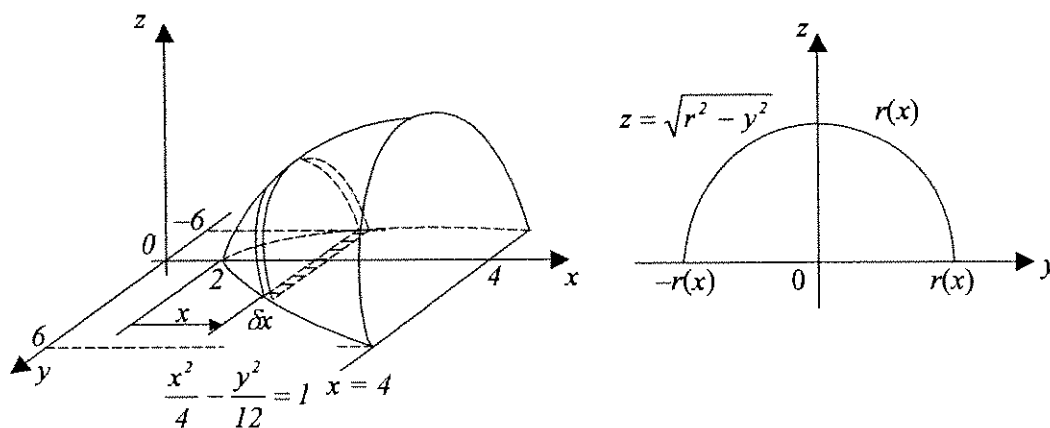
$\therefore$ The volume of the solid is $\pi(4 - e)$ cubic units. »

Question 6

b)

The base of a particular solid is the region bounded by the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$ between its vertex $(2, 0)$ and the corresponding latus rectum. Every cross-section perpendicular to the major axis is a semicircle with diameter in the base of the solid.

- Find the equation of the latus rectum.
- Find the volume of the solid.

SOLUTION:

The latus rectum of the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$ is the line $x = 4$.

The slice is a semicircle with radius r , area of cross-section A and thickness δx .

$$A(x) = \frac{\pi r^2(x)}{2},$$

$$r(x) = \sqrt{12} \cdot \sqrt{\frac{x^2}{4} - 1}$$

$$\therefore A(x) = 6\pi\left(\frac{x^2}{4} - 1\right)$$

$$\delta V = A(x)\delta x = 6\pi\left(\frac{x^2}{4} - 1\right)\delta x$$

The slice has volume

Then the volume of the solid is

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=2}^4 6\pi\left(\frac{x^2}{4} - 1\right)\delta x = 6\pi \int_2^4 \left(\frac{x^2}{4} - 1\right) dx$$

$$= 6\pi \left(\frac{x^3}{4 \cdot 3} - x \right) \Big|_2^4$$

$$= 16\pi$$

$\therefore$ The volume of the solid is 16π cubic units. »

Question 6

c)

The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$.

The chord PQ subtends a right angle at another point $R\left(cr, \frac{c}{r}\right)$ on the hyperbola.

Show that the normal at R is parallel to PQ .

SOLUTION:

$$xy = c^2$$

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = \frac{-y}{x} \text{ at } R\left(cr, \frac{c}{r}\right)$$

$$= \frac{-1}{r^2}$$

Hence, gradient of the normal at $R = r^2$

Let gradient of $RP = m_{RP}$

$$m_{RP} = \frac{\frac{c}{r} - \frac{c}{p}}{cr - cp}$$

$$= \frac{-1}{rp}$$

$$\text{Similarly } m_{RQ} = \frac{-1}{rq} \text{ and } m_{PQ} = \frac{-1}{pq}$$

$$\text{Now, } m_{RP} \times m_{RQ} = -1 \quad (\because \angle PRQ = 90^\circ)$$

$$\therefore \frac{1}{r^2 pq} = -1$$

$$r^2 = \frac{-1}{pq}$$

Hence, gradient of the normal at $R = m_{PQ}$

$\therefore$ Normal at R is parallel to PQ .

Q7b) SOLUTION

If $ax^3 + bx^2 + d = 0$ has a double root, show that $27a^2d + 4b^3 = 0$.

$$P(x) = ax^3 + bx^2 + d,$$

$$P'(x) = 3ax^2 + 2bx,$$

$$P''(x) = 6ax + 2b.$$

$P'(0) = 0$, $P'(-\frac{2b}{3a}) = 0$. Hence, both 0 and $-\frac{2b}{3a}$ can be a double root of $P(x) = 0$.

Let 0 be a double root.

Hence $P(0) = 0$, $d = 0 \Rightarrow$ if $27a^2d + 4b^3 = 0$, then $b = 0 \Rightarrow P(x) = ax^3$ and 0 is a triple root. Thus if 0 is a double root, then $27a^2d + 4b^3 \neq 0$.

Let $-\frac{2b}{3a}$ be a double root of $P(x) = 0$.

Hence

$$P(-\frac{2b}{3a}) = 0$$

$$a(-\frac{2b}{3a})^3 + b(-\frac{2b}{3a})^2 + d = 0$$

$$27a^2d + 4b^3 = 0$$

Question 8 (15 Marks)

a)

A coin is tossed six times. What is the probability that there will be more tails on the first three of the six throws than on the last three throws?

SOLUTION

3 outcomes: Equal tails, more tails or less tails.

$$P(\text{equal tails}) = P(1H) + P(2H) + P(3H) + P(0H)$$

$$= 9\left(\frac{1}{2}\right)^6 + 9\left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6$$

$$= \frac{20}{64}$$

$$P(\text{More tails in 1}^{\text{st}} 3 \text{ throws})$$

$$= \frac{1}{2} \left(1 - \frac{20}{64} \right)$$

$$= \frac{11}{32}$$

b)

If m points are taken on a straight line and n points on a parallel line, how many triangles can be drawn each having its vertices at 3 of the given points?

SOLUTION

Number of triangles

$$= m^n C_2 + n^m C_2$$

$$= \frac{1}{2} mn(m+n-2)$$

Question 8c)

(i)

$$\text{Show that } (1-x^2)^{\frac{n-3}{2}} - (1-x^2)^{\frac{n-1}{2}} = x^2 (1-x^2)^{\frac{n-3}{2}}.$$

SOLUTION

$$\begin{aligned} (1-x^2)^{\frac{n-3}{2}} - (1-x^2)^{\frac{n-1}{2}} &= x^2 (1-x^2)^{\frac{n-3}{2}} \\ &= (1-x^2)^{\frac{n-3}{2}} \left[1 - (1-x^2)^{\frac{2}{2}} \right] \\ &= x^2 (1-x^2)^{\frac{n-3}{2}} \end{aligned}$$

(ii)

SOLUTION

Using Integration by parts;

$$\begin{aligned} I_n &= \int_0^1 (1-x^2)^{\frac{n-1}{2}} \frac{d(x)}{dx} dx \\ &= \left[x(1-x^2)^{\frac{n-1}{2}} \right]_0^1 - \frac{n-1}{2} \int_0^1 -2x^2 (1-x^2)^{\frac{n-3}{2}} dx \end{aligned}$$

$$\therefore I_n = (n-1) \int_0^1 x^2 (1-x^2)^{\frac{n-3}{2}} dx$$

now from (c)i

$$I_n = (n-1) \int_0^1 x^2 (1-x^2)^{\frac{n-3}{2}} dx$$

$$I_n = (n-1) \int_0^1 (1-x^2)^{\frac{n-3}{2}} dx - (n-1) \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx$$

$$I_n = (n-1)I_{n-2} - (n-1)I_n$$

$$\therefore (n-1)I_n + I_n = (n-1)I_{n-2}$$

$$\text{Let } \therefore nI_n = (n-1)I_{n-2}$$

Question 8 (iii)

Let $J_n = nI_n \cdot J_{n-1}$ for $n = 1, 2, 3, \dots$.

By using mathematical induction, prove that

$$J_n = \frac{\pi}{2} \text{ for } n = 2, 3, \dots$$

SOLUTION

Test for $n=2$

$$J_2 = 2I_2 \cdot J_{2-1}$$

$$= I_0 \cdot I_1$$

$$I_2 = \int_0^1 (1-x^2)^{\frac{-1}{2}} dx \cdot \int_0^1 (1-x^2)^0 dx$$

$$= \int_0^1 (1-x^2)^{\frac{-1}{2}} dx$$

$$= [\sin^{-1} x]_0^1$$

$$= \frac{\pi}{2}$$

Assume true for $n=k$ ie $J_k = kI_k \cdot J_{k-1} = \frac{\pi}{2}$

Test for $n=k+1$

Now from $J_n = nI_n \cdot J_{n-1}$

$$J_{k+1} = (k+1)I_{k+1} \cdot J_k$$

And as $nI_n = (n-1)I_{n-2}$

therefore $(k+1)I_{k+1} = kI_{k-1}$

$$J_{k+1} = kI_{k-1} \cdot J_k$$

$$= \frac{\pi}{2}$$

Hence by Mathematical Induction

$$J_n = \frac{\pi}{2} \text{ for } n = 1, 2, 3, \dots$$

(iv)

Briefly explain why $0 < I_n < I_{n-1}$ for $n=1, 2, 3, \dots$

SOLUTION

$$I_n = \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx > 0 \text{ clearly!}$$

$$I_{n-1} = \int_0^1 (1-x^2)^{\frac{n-3}{2}} dx$$

$$\begin{aligned} \therefore I_n - I_{n-1} &= \int_0^1 \left[(1-x^2)^{\frac{n-1}{2}} - (1-x^2)^{\frac{n-3}{2}} \right] dx \\ &= \int_0^1 (1-x^2)^{\frac{n-3}{2}} \left[\sqrt{1-x^2} - 1 \right] dx \end{aligned}$$

$$\text{for } 0 \leq x \leq 1$$

$$0 \leq \sqrt{1-x^2} \leq 1$$

$$\therefore -1 \leq \sqrt{1-x^2} - 1 \leq 0$$

$$\text{Hence } I_n - I_{n-1} < 0$$

$$\therefore 0 < I_n < I_{n-1}$$

END OF PAPER